Review changelog: batch p4p5 (04-react-architecture 01-27, 05-design-patterns 01-26)

Date: 2026-10-22. Sandbox: React/ReactDOM 19.3.1, @types/react 19.3, TypeScript 5.9.3, Vitest 3, jsdom, RTL, TanStack Query 5.104, Zustand 5.0.16, XState 5.33, MSW 2.15. Extra scratch installs (in `tools/sandbox/scratch/p4p5/lint*`): ESLint 9.39.5 and 10.13.0, typescript-eslint 8.72, dependency-cruiser 18.7.0, eslint-plugin-boundaries 7.2.1, react-router 8.5, react-error-boundary. Scratch files: `tools/sandbox/scratch/p4p5/`.

Method: (1) extracted all 824 ts/tsx fenced blocks from the 53 lessons and ran `tsc --strict` on them (fragments produce many "cannot find name" errors; I triaged only the real type errors and re-ran key code assembled into complete files); (2) executed ~25 Vitest/RTL/node experiments for the stated predictions; (3) fetched react.dev (forwardRef, createContext, useEffectEvent, useActionState, createRoot, Component, upgrade guide, 19 / 19.2 posts), feature-sliced.design (layers, slices-segments, public-api, with-nextjs), jsboundaries.dev, mswjs.io migration, TS 6.0 announcement, designtokens.org, MDN, Node docs; (4) HTTP-checked all 245 external URLs (all 200 except bot-blocked dl.acm.org/medium.com 403, placeholder domains, and react-typescript-cheatsheet.netlify.app which did not respond). Not every one of the ~480 original warning markers was individually researched: see Unresolved.

Corrections (things that were wrong or would teach something false)

Lesson	Claim	How checked	Result / change
05/19 polymorphic §5.8	Render-prop `<a {...p} />` with `p` typed `ComponentPropsWithRef<'button'>`, "the type system will not stop you"	tsc 5.9	WRONG: it is a compile error (button handler/ref types vs anchor). Rewrote the example to `HTMLAttributes<HTMLElement>`, verified compiling, and corrected the prose.
05/19 §5.5	`<Button href type="submit">` is an ERROR	tsc	WRONG: `type` is a valid `<a>` attribute. Replaced with `formAction` (errors as claimed; `disabled` also errors).
05/19 §2	`JSX.Element` in signature	@types/react 19 has no global `JSX`	Changed to `React.JSX.Element`. `React.ElementRef` is `@deprecated` (use `ComponentRef`): updated table.
05/19 §5.3, Box type tests	canonical `as` typing, `ref` mismatch / `href` on span / `size` own-prop wins / Box tests	tsc with `@ts-expect-error` assertions	CONFIRMED (all assertions fire).
05/05 compound `Tabs`	`TabsList` spreads `{...props}` after its handler so a consumer `onKeyDown` silently disables arrow-key navigation	RTL + user-event	BUG confirmed (selected tab did not change). Fixed: consumer handler runs first, ours last; re-verified. Predict (state lost on tab switch), FormField render count (2; 4 in StrictMode) CONFIRMED.
05/18 headless tabs hook	object literal returns `'data-state'` typed as `ButtonHTMLAttributes`	tsc: TS2353	Return type now includes `'data-state'`.
05/15 registry	`components as Record<string, ...>`	tsc: TS2352	`as unknown as`.
04/17 event bus	`(handlers[name] ??= new Set()) as Set<Handler<K>>`	tsc: TS2322	Added `as never` with explanation.
useEffectEvent (04/01, 04/05, 04/07, 05/06 x2, 05/14, 05/22, 05/23)	"verify status", "stable callback" use, "in React 19"	react.dev/reference/react/useEffectEvent	Stable since React 19.2; function has deliberately unstable identity and must not be passed to children/hooks or put in deps; so it does NOT solve "stable callback for memoised child / returned from a hook". Fixed wording in 8 places.
04/17 §B3 and closure caveat	hook-level `useMutation` `onSuccess` captures cart "at the render of the click"	executed TanStack Query 5.104	WRONG for v5: `onSuccess` sees the LATEST render's closure (cart=2 after mid-request change). Corrected both places.

Lesson	Claim	How checked	Result / change
04/25 §5.1	`tseslint.config()`	typescript-eslint d.ts	Deprecated in favour of ESLint's `defineConfig` (`eslint/config`). Switched; verified on ESLint 9.39 and 10.13. Also noted the TS parser is needed for real .ts files.
04/25 §5.2	`eslint-plugin-boundaries` `element-types`, `rules`, `${from.x}`	executed v7.2.1	v7 deprecates all of these (`boundaries/dependencies`, `policies`, object selectors, `{{ }}` templates). Labelled the block as v5 API and added a v7 version that I ran (flags cross-feature import; needs import resolver).
04/25 §5.3	"`src/shared/**` may not lint .ts"	ESLint 9.39.5	CONFIRMED (".ts ignored: no matching configuration"); wording made precise. Override-replacement semantics, canary `lintText` test, dependency-cruiser config (all 5 rules fire; `$1` in `to.path` works; `--baseline` / `--ignore-known` exist): CONFIRMED.
04/26 D-key	"`!` negation not reliable in no-restricted-imports"	ESLint 9.39.5	Negation works but is a per-block (not per-feature) exception; reworded. `allowTypeImports` works on core rule too.
04/06 §3.2	tsconfig `baseUrl` + `paths`	TS 6.0 announcement	`baseUrl` deprecated in TS 6.0, unsupported in 7.0: removed from the example, note added.
04/06 §5.1	Zustand 5 selector returning new array	executed	CONFIRMED ("getSnapshot should be cached", "Maximum update depth exceeded"; `useShallow` fixes). Marker replaced by measured result.
04/24	React 19 removals table	upgrade guide	CONFIRMED all (propTypes, function defaultProps, legacy context, string refs, render/hydrate/unmountComponentAtNode, findDOMNode). Markers removed.
05/22	`useLayoutEffect` warns on server	`renderToString` on 19.3.1	No warning in 19.3: reworded (warning existed in 18). `require(esm)`: Node 22.12/20.19.
05/05	`useId` format	executed	19.3.1 gives `_r_0_` (client) / `_R_0_` (server); noted. Fragment `cloneElement` warning text confirmed.
05/22	"forwardRef deprecated for new code"	react.dev	Reworded to what react.dev says ("will be deprecated in a future release").
04/20, 04/22, 04/27, 04/09	DTCG spec, Turborepo `boundaries` / `$schema`, `@scope`, FSD items	web	DTCG 2025.10 first stable; `turbo boundaries` exists; turbo schema URL moved (turborepo.dev); `@scope` Baseline 2026; FSD claims below.
05/19 §bib	react-typescript-cheatsheet site unreachable	curl	Link now points to the GitHub repo.

Confirmed (no change beyond removing the marker)

- React 19 `ref` as prop; `forwardRef` "will be deprecated in a future release" (react.dev/reference/react/forwardRef); `<Context value>` provider, `.Provider` legacy / "will be deprecated" (react.dev createContext + React 19 post); `use(Context)`; ref callback cleanup (executed: cleanup runs, callback not called with null); `element.ref` access logs "Accessing element.ref was removed in React 19" (executed).
- Error boundaries: classes only (react.dev Component); `onCaughtError/onUncaughtError/onRecoverableError` (createRoot ref); throws in `useEffect` / `useLayoutEffect` reach boundaries (executed). Measured order on 19.3.1: gDSFE, fallback render,

onCaughtError, componentDidCatch; render retried several times (added as non-contractual note).
- Predictions executed and matching: reducer batching (3, 1 render), StrictMode reducer double call, Provider B1 (Inner 1 re-render), Sidebar 0, controlled Stepper logs 4,4, effect cleanup order `cleanup 0, effect 1`, assessment P1 (2), P2 (2 vs 1), P3 warning, P4 (max update depth), P5, P6 (`open, open, close`), D1 (track button only, linkRef null), form action resets uncontrolled input, `renderToString` throws for portals, `Children.count(<>..</>) === 1`.
- Typechecked OK: compound Tabs, FormField, polymorphic `as`, button union, Box type tests, Slot + mergeRefs (also runtime), reducer + `useReducer`, XState v5 machine (`setup().createMachine`, ran: `submitting:1 > failure:1 > submitting:2 > success:2`), generic table machine helper (F1), react-router 8 snippets, TanStack Query v5 exports (`QueryErrorResetBoundary`, `useQueryErrorResetBoundary`, `isInvalidated`), MSW 2 handler pattern, `useActionState` semantics (queued, errors to boundary).
- Feature-Sliced Design (v2.1 docs): layers and order, deprecated `processes`, import rule, app/shared exception, segments, `@x` notation (`entities/A/@x/B`), Next.js guide (prefixed `_app/_pages`), Steiger exists (0.8.x). Removed ⚠ and rewrote the header note.

Version notes added (newer majors than lesson targets)

MSW 3.0 (imports move to `msw/http`, Node 22) in 04/14; React Router 8.x in 05/17; ESLint 10 and TS 6/7 in 04/25 and 04/06; React Compiler 1.0 (Oct 2025) in 05/01, 05/02, 05/06, 05/10, 05/23; `<Activity>` shipped in 19.2 (05/05).

Unresolved / not verified

- About 270 of the original ~520 `⚠` markers remain: they are generic hedges about framework/RSC details (Next.js, React Router framework mode, TanStack Start), bundler/tool options (Vite, tsup, Rollup, Nx, changesets, pnpm flags, Style Dictionary, Storybook, openapi generators, API Extractor), CODEOWNERS platform quirks, MUI/Radix package paths, browser-support floors, Web Locks/BroadcastChannel, TC39 Observable. I did not research each; they stay as honest hedges.
- Throw inside an effect *cleanup* reaching a boundary: test failed to run cleanly, left unverified (stated as untested in 04/19).
- Redux/Immer/RTK, Pact, Radix `asChild` internals, Zod-specific API shapes: not executed beyond type-level import resolution.
- TanStack Router snippet (04/18) needs a generated route tree to typecheck; not verified.
- dl.acm.org and medium.com links return 403 to bots (probably fine); react-typescript-cheatsheet.netlify.app unreachable (replaced).
- Section-by-section reading of all 3.4 MB was not done; coverage is by claim type (grep of risky terms, ⚠ markers, all code blocks via tsc, all "Predict" items with concrete outputs in key lessons).